ICAD RT-5 · Set 2 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
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7	a	b	c	d
8	a	b	c	d
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16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
22	
23	
24	
25	

Chemistry

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
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6	a	b	c	d
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16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

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25	

Mathematics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
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18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
22	
23	
24	
25	

Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 2

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

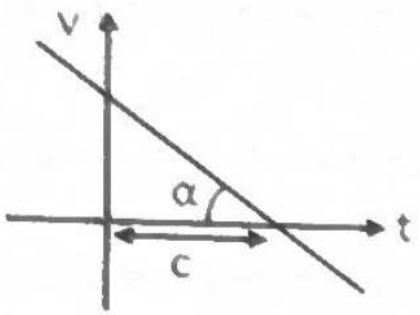
PART A · Q1–20 · SINGLE CORRECT

1. P2L4VSuggQ2 The minimum value of $f(x) = x^4 - x^2 - 2x + 6$ is (a) 6 (b) 4 (c) 8 (d) None

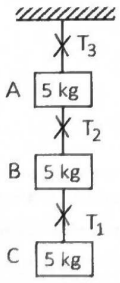
2. P3L5VSuggQ2 A and B are two vectors given by $A = 2\hat{i} + 3\hat{j}$ and $B = \hat{i} + \hat{j}$. The magnitude of the component of B along A is (a) $5/\sqrt{2}$ (b) $3/\sqrt{2}$ (c) $5/\sqrt{13}$ (d) $2/\sqrt{13}$

3. P4L3VSuggQ4 The ratio of distances traversed in successive intervals of time when a body falls freely under gravity from a certain height is: (a) 1:2:3 (b) 1:5:9 (c) 1:3:5 (d) $\sqrt{1}:\sqrt{2}:\sqrt{3}$

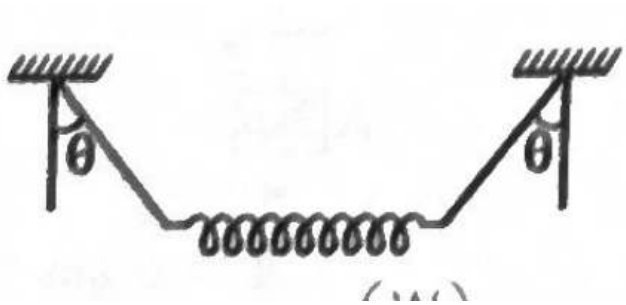
4. P4L6V5Q5 Find expression for velocity as a function of time for the given graph (v–t graph: a straight line sloping downward that cuts the t-axis at t = c, making an angle α with the t-axis). (a) $v = \tan\alpha(c + t)$ (b) $v = \tan\alpha(-c + t)$ (c) $v = \tan\alpha(c - t)$ (d) None of these



5. P5L2V5Q10 Three blocks A, B and C, each of mass 5 kg, hang one below the other from a ceiling (A on top, C at the bottom): T_3 is the tension in the string from the ceiling to A, T_2 between A and B, T_1 between B and C (see figure). The value of T_3/T_1 is: (a) 1 (b) 2 (c) 3 (d) 3/2

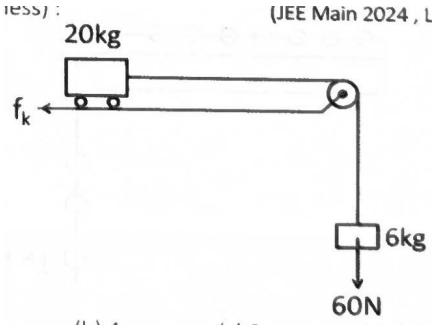


6. P6L5V6Q8 A stiff horizontal spring of weight W and force constant k is suspended in a horizontal position by two light strings attached to its two ends. Each string makes an angle θ with the vertical. The extension of the spring is: (a) (W/4k) tanθ (b) (W/2k) tanθ (c) (W/4k) sinθ (d) zero

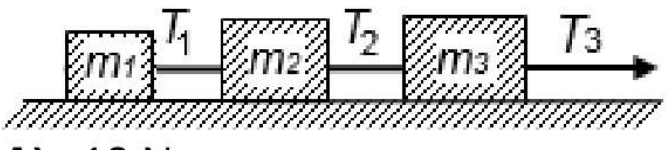


7. P6L7V5Q9 A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (where $m_2 > m_1$). If the acceleration of the system is $g/\sqrt{2}$, then the ratio of the masses m_1/m_2 is: (a) $(1+\sqrt{5})/(\sqrt{5}-1)$ (b) $(\sqrt{2}-1)/(\sqrt{2}+1)$ (c) $(1+\sqrt{5})/(\sqrt{2}-1)$ (d) $(\sqrt{3}+1)/(\sqrt{2}-1)$

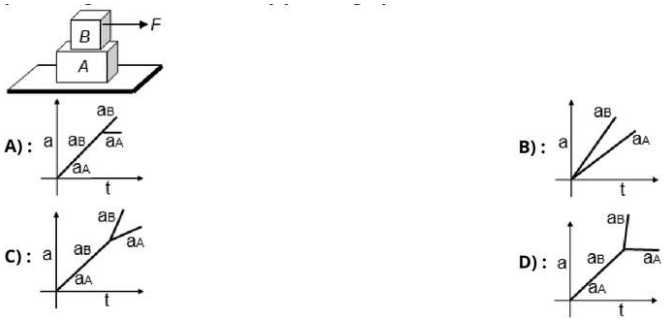
8. P6SBVPYQQ3 Consider a block and trolley system as shown in figure (a 20 kg trolley on a horizontal surface, joined by a string over a pulley at the table edge to a hanging 6 kg block of weight 60 N). If the coefficient of kinetic friction between the trolley and the surface is 0.04, the acceleration of the system in $m\ s^{-2}$ is: (Consider that the string is massless and unstretchable and the pulley is also massless and frictionless; $g = 10\ m/s^2$) (a) 3 (b) 4 (c) 2 (d) 1.2



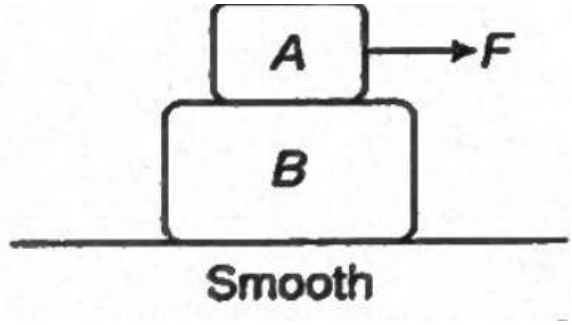
9. P6XVRT4Q7 Three blocks are connected as shown in the figure, on a horizontal frictionless table. If $m_1 = 1$ kg, $m_2 = 8$ kg, $m_3 = 27$ kg and $T_3 = 36$ N, T_2 will be : (a) 18 N (b) 9 N (c) 3.375 N (d) 1.75 N



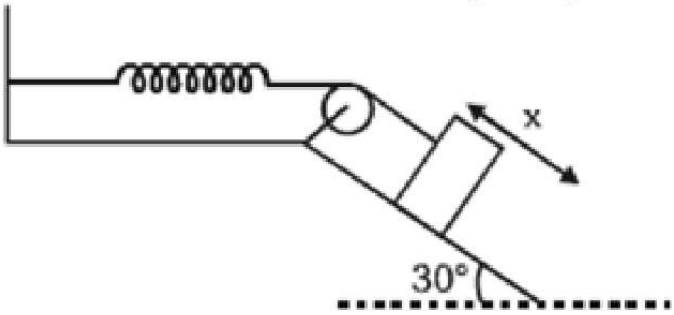
10. P7XVRT3Q14 A block B is placed on block A. The mass of block B is less than the mass of block A. Friction exists between the blocks, whereas the ground on which the block A is placed is taken to be smooth. A horizontal force F, increasing linearly with time, begins to act on B. The accelerations a_A and a_B of blocks A and B respectively are plotted against t. The correctly plotted graph is (a)–(d): the four a–t graphs shown in the figure



11. P8L1V5Q8 (one or more options may be correct) Block A slips over block B during motion (A is on top of B, a horizontal force F is applied to A, and B rests on a smooth floor), then: (a) the friction between the blocks is kinetic friction (b) the work done by friction on block B is positive (c) the work done by friction on block A is negative (d) heat loss takes place



12. P8XVCAT5Q13 A non-Hookian spring has force $F = -kx^2$ where k is the spring constant and x is the displacement from its unstretched position. For the system shown, a mass m connected to an unstretched spring initially at rest, how far does the spring extend before the system momentarily comes to rest? Assume that all surfaces are frictionless and that the pulley is frictionless as well. (a) (3mg/2k)^(1/2) (b) (mg/k)^(1/2) (c) (2mg/k)^(1/2) (d) (√3mg/2k)^(1/2)



13. P9L1VSuggQ1 A body initially at rest explodes into three fragments. Two of the fragments have equal mass and fly off with equal speeds in opposite directions. The third fragment will: (a) remain at rest (b) start moving in the direction of either fragment (c) move perpendicular to the direction of motion of the other two (d) move with zero momentum

14. P9L2V6Q4 A body of mass M moving at speed v_0 collides elastically with a mass m at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M. The largest possible value of the ratio M/m, for which the angles θ_1 and θ_2 will be equal, is (a) 4 (b) 1 (c) 3 (d) 2

15. P9SBVPYQQ9 A bag of sand of mass 9.8 kg is suspended by a rope. A bullet of 200 g travelling with speed 10 m s^{-1} gets embedded in it, then loss of kinetic energy will be (a) 4.9 J (b) 9.8 J (c) 14.7 J (d) 19.6 J

16. P10L4VSuggQ1 In a vertical circular motion, when is the kinetic energy maximum? (a) At the top of the circle (b) At the midpoint (c) At the bottom of the circle (d) It remains constant

17. P10SBVPYQQ14 A car is moving with a constant speed of 20 m s −1 on a circular horizontal track of radius 40 m. A bob is suspended from the roof of the car by a massless string. The angle made by the string with the vertical will be: (g = 10 m s −2) (a) π/6 (b) π/2 (c) π/4 (d) π/3

18. P10XVCAT6Q16 A car moves at constant speed on the road as shown in figure. The normal force by the road on the car is N_A and N_B when it is at the points A and B respectively. (a) N_A = N_B (b) N_A > N_B (c) N_A < N_B (d) insufficient information to decide the relation of N_A and N_B



19. P11L1V2Q4 Two bodies of mass 1 kg and 3 kg have position vector $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$, respectively. The centre of mass of this system has a position vector (a) $-\hat{i} + \hat{j} + \hat{k}$ (b) $-2\hat{i} + 2\hat{k}$ (c) $-2\hat{i} - \hat{j} + \hat{k}$ (d) $2\hat{i} - \hat{j} - 2\hat{k}$

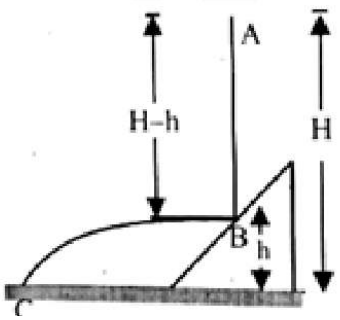
20. P11L2VSuggQ1 What is the expression for mass per unit length (λ) of a uniform rod of mass M and length L? (a) λ = M×L (b) λ = L/M (c) λ = M/L (d) λ = 1/(ML)

PART B · Q21–25 · NUMERICAL VALUE

21. P4L11VSuggQ2 Rain falling vertically downwards at 4 km/h. A girl moves on a straight road with velocity 3 km/h. Apparent velocity of rain w.r.t. girl?

22. P4XVCAT3Q23 The velocity of a projectile when it is at the greatest height is √(2/5) times its velocity when it is at half of its greatest height. Determine its angle of projection in degree.

23. P4XVRT3Q24 A body falling freely from a given height H hits an inclined plane in its path at a height h. As a result of this impact, the direction of the velocity of the body becomes horizontal. For what value of H/h will the body take maximum time to reach the ground?



24. P8L1VSuggQ2 A body constrained to move along the z-axis of a co-ordinate system is subjected to a constant force F□ = −î + 2ĵ + 3k̂ N, where î, ĵ, k̂ are unit vectors along x, y and z. What is the work done by this force in moving the body a distance of 4 m along the z-axis?

25. P11L1V6Q1 Three identical spheres each of mass M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 3 m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be √x m. The value of x is

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L1V5Q5 Two identical mercury barometers are placed at sites A and B, both exposed to the same atmospheric pressure. At site A, the temperature is 0 °C, while at site B, the temperature is higher. Assume the barometric relation P = dgh and that the density of mercury decreases with increase in temperature. Which of the following correctly describes the relative heights of mercury columns at A and B? (a) h_B = h_A (b) h_B > h_A (c) h_B < h_A (d) Insufficient data to compare

2. C2L1V5Q5 Rutherford found that about one in 20,000 α-particles was deflected back through angles > 90°. Which statement agrees with this observation? (a) A very heavy and dense body exists in the atom occupying a small volume. (b) Positive charge is uniformly spread throughout the atom. (c) Back-scattering occurs due to electrons. (d) α-Particles convert into protons on collision.

3. C2L7V5Q5 The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are: (a) Lyman and Paschen (b) Balmer and Brackett (c) Brackett and Pfund (d) Paschen and Pfund

4. C3L8VSuggQ5 Two samples of HCl of 1.0 M and 0.25 M are mixed. Find volumes of these samples taken in order to prepare 0.75 M HCl solution. Assume no water is added. (I) 20 mL, 10 mL (II) 100 mL, 50 mL (III) 40 mL, 20 mL (IV) 50 mL, 25 mL (a) I, II, IV (b) I, II (c) II, III, IV (d) I, II, III, IV

5. C3SBVPYQQ14 Combustion of glucose ($C_6H_{12}O_6$) produces CO_2 and water. The amount of oxygen (in g) required for the complete combustion of 900 g of glucose is: [Molar mass of glucose in g mol^{-1} = 180] (a) 480 (b) 800 (c) 960 (d) 32

6. C4L4V6Q2 Arrange the following in increasing order of the property indicated: (A) Mg, Al, Si and Na (ionisation enthalpy) (B) C, N, O and F (2nd ionisation enthalpy) (a) Na < Al < Mg < Si ; C < N < F < O (b) Na < Al < Mg < Si ; C < F < N < O (c) Al < Na < Mg < Si ; C < N < F < O (d) Mg < Na < Al < Si ; F < O < N < C

7. C4L5V4Q3 Electron affinity of noble gases is (a) very high (b) low (c) almost approaching zero (d) high

8. C4SBVPYQQ47 The electron affinity value are negative for: A. Be → Be^- B. N → N^- C. O → O 2− D. Na → Na^- E. Al → Al^-. Choose the most appropriate answer from the options given below: (a) D and E only (b) A, B and C only (c) A and D only (d) A, B, D and E only

9. C4SBVPYQQ57 The correct order of electronegativity for given elements is (a) P > Br > C > At (b) Br > P > At > C (c) Br > C > At > P (d) C > P > At > Br

10. C5L1VSuggQ5 A bond is most likely to form when: (a) Repulsive forces dominate and energy increases (b) Attractive forces dominate and energy decreases (c) Both attractive and repulsive forces are equal (d) Electrons cannot redistribute between atoms

11. C5L3VSuggQ4 Why is head-on overlap stronger than lateral overlap? (a) Because p-orbitals cannot undergo lateral overlap (b) Because electron cloud concentration is greater along the internuclear axis (c) Because energy of p-orbitals decreases during lateral overlap (d) Because s-orbitals cannot overlap head-on

12. C5L10VSuggQ2 Which compound has the highest boiling point due to hydrogen bonding? (a) H_2S (b) HBr (c) H_2O (d) PH_3

13. C5XVCAT5Q38 Why is BF_3 less acidic than BCl_3 despite fluorine being more electronegative than chlorine? (a) Because B–Cl bonds are stronger than B–F bonds (b) Due to the smaller atomic size of fluorine (c) Because fluorine can form strong pπ–pπ back bonds with boron (d) Because chlorine is more polarizable than fluorine

14. C6L1V3Q3 If any one macroscopic property of a system changes, what happens to the system? (a) It remains in the same state (b) It reaches equilibrium (c) Its state changes (d) It becomes isolated

15. C6L9V3Q2 Given: Heat of vaporisation = 40.6 kJ/mol; how much energy needed to vaporize 2 mol H 2 O? (a) 40.6 kJ (b) 81.2 kJ (c) 20.3 kJ (d) 101.5 kJ

16. C6L9V4Q3 The heat evolved during neutralization is maximum in the reaction of: (a) NH 4 OH and CH 3 COOH (b) NH 4 OH and HCl (c) NaOH and CH 3 COOH (d) NaOH and HCl

17. C6L10V5Q10 An ideal gas expands in volume from 10 −3 m 3 to 10 −2 m 3 at 300 K against a constant pressure of 10 5 N m −2 . The work done is: (a) 900 kJ (b) − 900 kJ (c) 270 kJ (d) − 900 J

18. C6L11V3Q2 Calculate ΔG at 500 K for a reaction where ΔH = 100 kJ/mol and ΔS = 250 J/mol·K. Then, determine if the reaction is spontaneous. (a) 25 kJ, non Spontaneous (b) −25 kJ, Spontaneous (c) 0 kJ, Equilibrium (d) −125 kJ, Spontaneous

19. C6L11V4Q4 Identify the correct statement regarding a spontaneous process: (a) Lowering of energy is the only criterion for spontaneity (b) For a spontaneous process in an isolated system, the change in entropy is positive (c) Endothermic process are never spontaneous (d) Exothermic process are always spontaneous

20. C6L12V3Q3 For a cell with n = 3 and E° = 0.85 V, calculate ΔG° (F = 96500 C mol^{-1}), then find the amount of work in kJ. (a) −245.9 kJ (b) −250.0 kJ (c) −246.1 kJ (d) −248.0 kJ

PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT1Q36 A 1 mol sample of N_2 diffuses through a small orifice at 0.8 atm in 38 s. Under identical temperature, 1 mol of a xenon–fluorine compound at 1.6 atm takes 57 s to diffuse through the same hole. Determine the molecular mass of the compound. (A) 208 (B) 228 (C) 252 (D) 283

22. C2XVRT1Q47 Photons of same energy strike two different samples of hydrogen atoms, one with each atom in the ground state and the other in a particular excited state of orbit number n. The photonic beams ionize the hydrogen atoms in both samples. If the difference in maximum kinetic energy of the emitted photoelectron from the two samples is 12.75 eV, then the value of n is ____.

23. C2XVRT4Q49 How many electrons can have the principal quantum number n = 4 and the spin quantum number m s = +1/2 in an atom?

24. C3XVCAT4Q41 100 mL of 0.1 M HCl is titrated with 0.2 N NaOH; the titration is stopped after adding 30 mL of NaOH. The rest is completed with 0.25 N KOH. The volume of KOH required is: (A) 70 mL (B) 32 mL (C) 35 mL (D) 16 mL

25. C5XVRT4Q48 Maximum number of atoms present linearly in the following molecule is H 2 C=C=CH–C≡C–C≡C–CH 3

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M2XVRT3Q66 Let n(U) = 700, n(A) = 200, n(B) = 300 and n(A ∩ B) = 100, then $n(A^c \cap B^c)$ equals : (a) 400 (b) 600 (c) 300 (d) 200

2. M3L1VSuggQ4 If tan θ = 3/4 and θ lies in the third quadrant, then cos θ = ? (a) 4/5 (b) −4/5 (c) 3/5 (d) −3/5

3. M4L2V3Q5 If f(x) = { $x^2 - 1$ if x ≤ 0 ; x − 1 if x > 0 }, then the value of f(−2) + f(3) is (a) 5 (b) 2 (c) 10 (d) None

4. M5AL3VSuggQ8 If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has real solutions for θ, then λ lies in the interval: [Sep. 02, 2020 (III)] (a) (−5/4, −1) (b) [−1, −1/2] (c) (−1/2, −1/4] (d) [−3/2, −5/4]

5. M5CXVRT4Q66 The product (cos x/2) · (cos x/4) · (cos x/8) · ... · (cos x/256) is equal to: (A) sin x / (128 sin(x/256)) (B) sin x / (256 sin(x/256)) (C) sin x / (128 sin(x/128)) (D) sin x / (512 sin(x/512))

6. M6L4V5Q8 If |2x + 3| − |2x − 5| = 6, then x is (a) 1 (b) 3 (c) 4 (d) 2

7. M6XVRT4Q67 If the natural domain of the function f(x) = √(log₂((x − 2)/(3 − x))) is [a, b), then find the value of (2a − b) is (a) −1 (b) 0 (c) 1 (d) 2

8. M7L2V5Q9 Which of the following, when simplified, reduces to unity (where a = 23 and b = 78)? (One or more options may be correct.) (one or more options may be correct) (a) log10 √(25^(1/log8 5) + 49^(1/log6 7)) (b) log_(1/2)(log₃ cos30° − log₃ sin30°) (c) ((5√3 + √50)(5 − √24))/(√75 − 5√2) (d) [(a³ − a − 2b − b²/a) / ((1 − √(1/a + b/a²))·(a + √(a + b)))] : [(a³ + a² + ab + a²b)/(a² − b²) + b/(a − b)]

9. M7XVCAT4Q66 If log10x + log10y = 2, x − y = 15 then: (a) (x, y) lies on the line y = 2x + 3 (b) (x, y) lies on $y^2 = 4x$ (c) (x, y) lies on x = 4y (d) (x, y) lies on 4x = y

10. M8AXVCAT5Q67 Let $a_1, a_2, \ldots, a_n$ be a given A.P. whose common difference is an integer and $S_n = a_1 + a_2 + \ldots + a_n$. If $a_1 = 1$, $a_n = 300$ and $15 \le n \le 50$, then the ordered pair (S_{n-4}, a_{n-4}) is equal to : (a) (2480, 249) (b) (2490, 249) (c) (2490, 248) (d) (2480, 248)

11. M8BL2V4Q4 The first term of a G.P. whose second term is 2 and sum to infinite is 8, will be (a) 6 (b) 3 (c) 4 (d) 1

12. M8CL1V4Q5 If a, b, c, d are in H.P., then ab + bc + cd is equal to (a) 3ad (b) (a + b)(c + d) (c) 3ac (d) None of these

13. M8DL1VSuggQ2 Find the sum of first 10 natural numbers. (a) 50 (b) 55 (c) 60 (d) 45

14. M8DL4V5Q9 The sum of (n − 1) terms of 1 + (1 + 3) + (1 + 3 + 5) + ... is (a) n(n+1)(2n+1)/6 (b) $n^2(n+1)/4$ (c) n(n−1)(2n−1)/6 (d) n^2

15. M8DL4VSuggQ1 What is the arithmetic mean (A.M.) between two numbers a and b? [Level-1] (a) √(ab) (b) (a + b)/2 (c) ab (d) 2ab/(a + b)

16. M9L2V2Q4 Find the modulus of z = 3 + 4i. (a) 5 (b) 7 (c) 25 (d) 6

17. M9L3V2Q1 The conjugate of (i) -2 + 3i (ii) 1 - i (iii) 4 (iv) 4i in order are (a) 2 - 3i, 1 + i, 4, -4i (b) +2 - 3i, 1 + i, 4, -4i (c) 2 - 3i, 1 + i, 4, 4i (d) -2 - 3i, 1 + i, 4, -4i

18. M9L5V1Q3 Which formula gives the modulus of z = x + iy? (a) $x^2 + y^2$ (b) $\sqrt{(x^2 + y^2)}$ (c) x + y (d) x - y

19. M9L6V2Q3 Let z_1, z_2 be two complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$. Then, (a) $\arg(z_1) = \arg(z_2)$ (b) $\arg(z_1) + \arg(z_2) = \pi/2$ (c) $|z_1| = |z_2|$ (d) $z_1 z_2 = 1$

20. M9L7V1Q2 The symbol ω is defined as: (a) 1 (b) e^(iπ) (c) (−1 + √3 i)/2 (d) (1 + √3 i)/2

PART B · Q21–25 · NUMERICAL VALUE

21. M4XVRT2Q64 If y = 2[x] + 3 and y = 3[x] − 1, then the value of [x + y], where [·] denotes the greatest integer function, is (A) 11 (B) 15 (C) 4 (D) 5

22. M7SBVPYQQ5 Let S={α : log₂(9^(2α−4)+13) − log₂((5/2)·3^(2α−4)+1)=2}. Then the maximum value of β for which the equation $x^2 - 2(\Sigma_{\alpha \in S}\alpha)^2 x + \Sigma_{\alpha \in S}(\alpha+1)^2\beta = 0$ has real roots, is ____

23. M8AL3VSuggQ5 A man saves Rs. 200 in each of the first three months of his service. In each of the subsequent months his savings increased by Rs. 40 more than the saving of immediately previous month. After how many months from the start of service, his total saving will be Rs. 11040.

24. M8BSBVPYQQ1 In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is R − (a, b), then $a^2 + b^2$ is equal to _____.

25. M8DSBVPYQQ7 Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\Sigma_{r=1}^{\infty} a_r / 2^r = 4$, then $4a_2$ is equal to _____